

4 Light–matter interaction and spectra

4.1 Time-dependent Hamiltonians

The TDSE is:

$$i\hbar \frac{\partial \Psi(t)}{\partial t} = \hat{H}(t) \Psi(t). \quad (4.1)$$

In this section we introduce time-dependent perturbation theory. We assume that the Hamiltonian can be written as a time-independent and a (possibly) time-dependent term, $\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t)$. It is used to develop a series of successive approximations to the evolving wave functions. Time-dependent perturbation theory is:

- Useful if dynamics under $\hat{H}_0$ are known
- Only recommended if $\hat{H}_1 \ll \hat{H}_0$
- derived either using standard perturbation theory or more directly from the interaction picture. We use the latter derivation here.

The propagator associated with $\hat{H}_0$ is given by

$$i\hbar \frac{\partial}{\partial t} \hat{U}_0(t, t_0) = \hat{H}_0 \hat{U}_0(t, t_0). \quad (4.2)$$

As before, the solution to this is $\hat{U}_0(t, t_0) = e^{-i\hat{H}_0(t-t_0)/\hbar}$.

The full propagator obeys the equation

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H}(t) \hat{U}(t, t_0). \quad (4.3)$$

If $\hat{H}(t)$ is time-dependent, the solution to this equation cannot be written down in a simple way. Nonetheless we can define the interaction-picture propagator as

$$\hat{U}_I(t, t_0) = \hat{U}_0(t, t_0)^\dagger \hat{U}(t, t_0). \quad (4.4)$$

Substituting $\hat{U} = \hat{U}_0 \hat{U}_I$ into the TDSE gives

$$i\hbar \frac{\partial}{\partial t} \hat{U}_I(t, t_0) = \hat{H}_1(t) \hat{U}_I(t, t_0) \quad (4.5)$$

where $\hat{H}_I(t) = e^{\frac{i\hat{H}_0(t-t_0)}{\hbar}} \hat{H}_1(t) e^{-\frac{i\hat{H}_0(t-t_0)}{\hbar}}.$

Exercise: derive this formula.

Equation (4.5) is solved by

$$\hat{U}_I(t, t_0) = 1 + \sum_{n=1}^{\infty} \hat{U}_I^{(n)}(t, t_0) \quad (4.6)$$

where

$$\hat{U}_I^{(n)}(t, t_0) = \frac{1}{(i\hbar)^n} \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau_{n-1} \cdots \int_{t_0}^{\tau_2} d\tau_1 \hat{H}_I(\tau_n) \hat{H}_I(\tau_{n-1}) \cdots \hat{H}_I(\tau_1) \quad (4.7)$$

with ordering $t > \tau_n > \tau_{n-1} > \dots > \tau_1 > t_0$.

Exercise: Show that this is a valid solution.

Therefore

$$\hat{U}(t, t_0) = \hat{U}_0(t, t_0) + \sum_{n=1}^{\infty} \hat{U}^{(n)}(t, t_0) \quad (4.8)$$

where

$$\begin{aligned} \hat{U}^{(n)}(t, t_0) &= \frac{1}{(i\hbar)^n} \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau_{n-1} \dots \int_{t_0}^{\tau_2} d\tau_1 \hat{U}_0(t, \tau_n) \hat{H}_1(\tau_n) \hat{U}_0(\tau_n, \tau_{n-1}) \\ &\quad \times \hat{H}_1(\tau_{n-1}) \dots \hat{U}_0(\tau_2, \tau_1) \hat{H}_1(\tau_1) \hat{U}_0(\tau_1, t_0). \end{aligned} \quad (4.9)$$

Substituting $\hat{U}_0(t, t') = e^{\frac{-i\hat{H}_0(t-t')}{\hbar}}$ gives

$$\begin{aligned} \psi^{(1)}(t) &= \frac{1}{i\hbar} \int_{t_0}^t dt' e^{\frac{-i\hat{H}_0(t-t')}{\hbar}} \hat{H}_1(t') e^{\frac{-i\hat{H}_0(t'-t_0)}{\hbar}} \psi(t_0) \\ \psi^{(2)}(t) &= \frac{1}{(i\hbar)^2} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' e^{\frac{-i\hat{H}_0(t-t')}{\hbar}} \hat{H}_1(t') e^{\frac{-i\hat{H}_0(t'-t'')}{\hbar}} \hat{H}_1(t'') e^{\frac{-i\hat{H}_0(t''-t_0)}{\hbar}} \psi(t_0). \end{aligned} \quad (4.10)$$

4.2 Interaction of light with matter

Here we make a couple of assumptions.

Dipole approximation: valid if wavelength of light is much greater than size of molecule

Optical domain: assume that $\hat{V}_{\text{el}}$ dominates over $\hat{V}_{\text{mag}}$.

We therefore introduce the light-matter interaction:

$$\hat{V}_{\text{el}}(t) = -\vec{\mu}_{\text{el}} \cdot \vec{\mathcal{E}}(t) \quad (4.11)$$

where $\vec{\mathcal{E}}(t)$ is the instantaneous electric field of the light. In other courses on EPR and NMR, you may have seen a similar expression for the magnetic interaction:

$$\hat{V}_{\text{mag}}(t) = -\vec{\mu}_{\text{mag}} \cdot \vec{B}(t). \quad (4.12)$$

In this course, we will only consider electric-field interactions and thus drop the subscript “el” in the following.

The (electric) dipole moment vector is defined in terms of the positions of all charges (including electrons and nuclei) in the molecule

$$\vec{\mu} = \sum_i Z_i \vec{R}_i - \sum_j \vec{r}_j. \quad (4.13)$$

With a continuous-wave monochromatic laser, one has the electric field

$$\vec{\mathcal{E}}(t) = \vec{\mathcal{E}}_0 \cos(\omega t) \quad (4.14)$$

and in the opposite limit of a short pulse at time t_0 :

$$\vec{\mathcal{E}}(t) \propto \delta(t - t_0). \quad (4.15)$$

In reality the fields are typically intermediate between these two limits.

4.3 Dipole correlation functions and infra-red spectra

Here, we will assume that the Born–Oppenheimer approximation holds and add a continuous-wave light field. Our total wavefunctions can be written as a product of an electronic part, $|\phi\rangle$, and a time-dependent nuclear part $|\chi(t)\rangle$. We integrate out all electronic variables to get the Born–Oppenheimer Hamiltonian $\hat{H}_0$ and a perturbing Hamiltonian

$$\hat{H}_1(t) = -\hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t), \quad (4.16)$$

where the dipole moment operator, which depends on nuclear positions only, is

$$\hat{\mu}(\vec{R}) = \langle \phi | \vec{\mu} | \phi \rangle_{\vec{r}} \quad (4.17)$$

and here the bracket indicates integration over electronic variables only.

We choose our initial wavefunction, $|\chi(t_0)\rangle$, to be a vibrational eigenstate on the ground electronic state and we wish to calculate how much population of this state is lost to another vibrational state. This will be proportional to the amount of light absorbed, which is what is measured in an IR spectrum.

The first-order perturbed nuclear wavefunction at time t is

$$|\chi^{(1)}(t)\rangle = \frac{-1}{i\hbar} \int_{t_0}^t dt' e^{-i\hat{H}_0(t-t')/\hbar} \hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t') e^{-i\hat{H}_0(t'-t_0)/\hbar} |\chi(t_0)\rangle \quad (4.18)$$

Integrating over the absolute-value squared of the first-order perturbed wavefunction gives

$$\begin{aligned} \langle \chi^{(1)}(t) | \chi^{(1)}(t) \rangle &= \frac{1}{\hbar^2} \int_{t_0}^t dt'' \int_{t_0}^t dt' \langle \chi(t_0) | e^{i\hat{H}_0(t''-t_0)/\hbar} \hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t'') e^{i\hat{H}_0(t-t'')/\hbar} \\ &\quad \times e^{-i\hat{H}_0(t-t')/\hbar} \hat{\mu} \cdot \vec{\mathcal{E}}_0 \cos(\omega t') e^{-i\hat{H}_0(t'-t_0)/\hbar} | \chi(t_0) \rangle \end{aligned} \quad (4.19)$$

where here the bracket indicates integration over nuclear variables. We can now take the limits that $t_0 \rightarrow -\infty$ and $t \rightarrow \infty$ and remove phase terms which anyway cancel to give the absorption spectrum as

$$I(\omega) \propto \langle \chi^{(1)}(\infty) | \chi^{(1)}(\infty) \rangle \quad (4.20)$$

$$\propto \frac{1}{\hbar^2} \iint_{-\infty}^{\infty} dt'' dt' \langle \chi(-\infty) | \hat{\mu} e^{i\hat{H}_0(t'-t'')/\hbar} \hat{\mu} e^{-i\hat{H}_0(t'-t'')/\hbar} | \chi(-\infty) \rangle \cos(\omega t'') \cos(\omega t') \quad (4.21)$$

Bearing in mind that

$$\cos(\omega t') \cos(\omega t'') = \frac{1}{2} [\cos \omega(t' - t'') + \cos \omega(t' + t'')] \quad (4.22)$$

we see that only the first term contributes, as the second averages out to 0. Therefore, changing variables again using $t = t' - t''$, we get

$$\langle \chi^{(1)}(\infty) | \chi^{(1)}(\infty) \rangle \propto \int_{-\infty}^{\infty} dt \langle \chi(-\infty) | \hat{\mu} e^{i\hat{H}_0 t/\hbar} \hat{\mu} e^{-i\hat{H}_0 t/\hbar} | \chi(-\infty) \rangle \cos(\omega t) \quad (4.23)$$

$$\propto \int_{-\infty}^{\infty} dt \langle \chi(-\infty) | \hat{\mu} \cdot \hat{\mu}(t) | \chi(-\infty) \rangle \cos(\omega t) \quad (4.24)$$

This is easily extended to treat an ensemble of initial states described by the density matrix $\hat{\rho}$ and gives the formula for the absorption as the Fourier (cosine) transform

$$I(\omega) \propto \int_{-\infty}^{\infty} C_{\mu\mu}(t) \cos(\omega t) dt \quad (4.25)$$

where the dipole-dipole correlation function is

$$C_{\mu\mu}(t) = \text{Tr}[\hat{\rho} \hat{\mu} \hat{\mu}(t)]. \quad (4.26)$$

In the simplest cases, we can expand the dipole operator as a Taylor series around the minimum-energy geometry, $\vec{R}_0$, as

$$\hat{\mu}(\vec{R}) = \mu(\vec{R}_0) + \nabla \mu(\vec{R}_0) \cdot (\vec{R} - \vec{R}_0) + \dots \quad (4.27)$$

Truncating this series after the first term gives a simple approximation. Then in equilibrium we have

$$C_{\mu\mu}(t) \approx [\nabla \mu(\vec{R}_0)]^2 \text{Tr}[\hat{\rho}_{\text{eq}} (\vec{R} - \vec{R}_0) \cdot (\vec{R}(t) - \vec{R}_0)] \quad (4.28)$$

because all other terms are identically zero. This therefore becomes the position-position correlation function. Note that you already computed this correlation function for the harmonic oscillator in the second lecture. It was proportional $\cos(\omega_0 t)$, where ω_0 is the frequency of the oscillator. By Fourier-transforming it, you can find the infra-red spectrum of the harmonic oscillator, which gives two delta functions, one at $\omega = \omega_0$ and the other at $\omega = -\omega_0$ corresponding to absorption and emission of light.

We can check that this approach based on correlation functions gives us the same spectrum as the time-independent approach that you are more familiar with. To do this, we evaluate the trace in equation (5.55) in the basis of (vibrational) eigenstates to get

$$C_{\mu\mu}(t) = \sum_{nm} P_n \langle n | \hat{\mu} | m \rangle \langle m | \hat{\mu}(t) | n \rangle \quad (4.29)$$

$$= \sum_{nm} P_n \mu_{nm} e^{iE_m t/\hbar} \mu_{mn} e^{-iE_n t/\hbar} \quad (4.30)$$

where $\mu_{nm} = \langle n | \hat{\mu} | m \rangle$ is the transition dipole moment. Taking the Fourier transform gives the spectrum

$$I(\omega) \propto \int_{-\infty}^{\infty} \sum_{nm} P_n |\mu_{nm}|^2 \left[e^{i(\omega_{nm} - \omega)t} + e^{i(\omega_{nm} + \omega)t} \right] dt \quad (4.31)$$

$$\propto \sum_{nm} P_n |\mu_{nm}|^2 [\delta(\omega_{nm} - \omega) + \delta(\omega_{nm} + \omega)] \quad (4.32)$$

where $\omega_{nm} = (E_m - E_n)/\hbar$. This is a stick spectrum of lines appearing at frequencies corresponding to the gaps between energy levels weighted by the Boltzman distribution of the initial state and the transition dipole moment squared. The first delta function describes absorption whereas the second describes stimulated emission.

4.4 Rabi oscillations in two-level system

In this section we allow the field strength to be strong, and thus cannot employ the perturbation approach.

We consider a two-level system described by a field-free Hamiltonian $\hat{H}_{\text{mol}}$ and a time-dependent interaction $\hat{V}(t)$ with an electromagnetic field:

$$\mathbf{H} = \begin{pmatrix} \hbar\omega_a & -\mu\epsilon \cos(\omega t) \\ -\mu\epsilon \cos(\omega t) & \hbar\omega_b \end{pmatrix}, \quad (4.33)$$

where $\mu\epsilon = \langle \psi_a | \hat{\mu} | \psi_b \rangle E_0$.

We use the interaction picture to write the wave function as

$$\psi(t) = a(t)e^{-i\omega_a t}\psi_a + b(t)e^{-i\omega_b t}\psi_b. \quad (4.34)$$

And therefore the Schrödinger equation is

$$\dot{a}(t) = \frac{i\mu\epsilon}{2\hbar} \left[e^{i(\omega - \omega_{ab})t} + e^{-i(\omega + \omega_{ab})t} \right] b(t) \quad (4.35a)$$

$$\dot{b}(t) = \frac{i\mu\epsilon}{2\hbar} \left[e^{-i(\omega - \omega_{ab})t} + e^{i(\omega + \omega_{ab})t} \right] a(t), \quad (4.35b)$$

where $\omega_{ab} = \omega_b - \omega_a$.

Only approximate solutions of this system of coupled differential equations are known analytically. We employ the “rotating wave approximation” which is valid if the frequency of the light, ω , is close to ω_{ab} , the energy gap. If the detuning $\Delta\omega = \omega - \omega_{ab}$ is small, i.e. $\Delta\omega \ll \omega_{ab}$ then one of the terms will oscillate very fast, and the other very slowly. We can ignore the fast oscillations as they will cancel out in the integration

Therefore within this approximation, we find

$$\dot{a}(t) = \frac{i\mu\epsilon}{2\hbar} e^{i\Delta\omega t} b(t) \quad (4.36a)$$

$$\dot{b}(t) = \frac{i\mu\epsilon}{2\hbar} e^{-i\Delta\omega t} a(t). \quad (4.36b)$$

These equations, along with the initial conditions $|a(0)| = 1$ and $b(0) = 0$, are solved by

$$a(t) = e^{i\Delta\omega t/2} \left(\cos(\Omega t/2) - i \frac{\Delta\omega}{\Omega} \sin(\Omega t/2) \right) \quad (4.37a)$$

$$b(t) = e^{-i\Delta\omega t/2} \frac{\mu\epsilon}{2\hbar\Omega} 2i \sin(\Omega t/2). \quad (4.37b)$$

Therefore for a state starting in a , the population of b goes like

$$|b(t)|^2 = \left(\frac{\mu\epsilon}{\hbar\Omega} \right)^2 \sin^2(\Omega t/2), \quad (4.38)$$

where $\Omega = \sqrt{\Delta\omega^2 + \mu^2\epsilon^2/\hbar^2}$.

- Note that the population dynamics are independent of detuning, $\Delta\omega$, at short times.
- At resonance, $\Delta\omega = 0$, it is possible to put all the population into the excited state if the laser pulse is turned off at $t = \pi/\Omega$.

4.5 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Time-dependent perturbation theory

- Tuckerman, Statistical Mechanics (Oxford), chapter 14
- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapter 9

Spectra and dipole correlation functions

- Heller, The Semiclassical Way to Dynamics and Spectroscopy, chapter 17
- May and Kühn, Charge and Energy Transfer Dynamics in Molecular Systems (Wiley, 2011)

Rabi oscillations

- Tannor, Introduction to quantum mechanics: a time-dependent perspective, chapter 15
- Grynberg, Aspect and Fabre, Introduction to quantum optics (Cambridge, 2010)